Bypass surgery

Heart information

Heart Foundation

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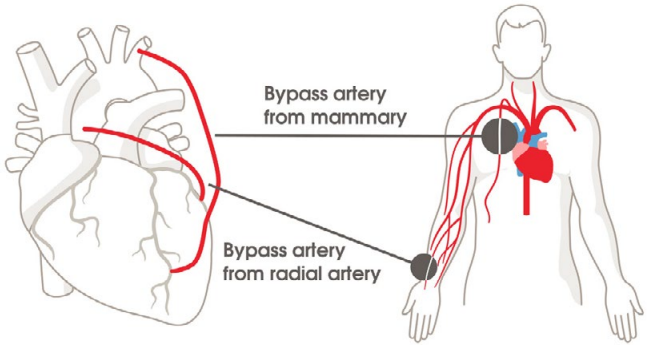
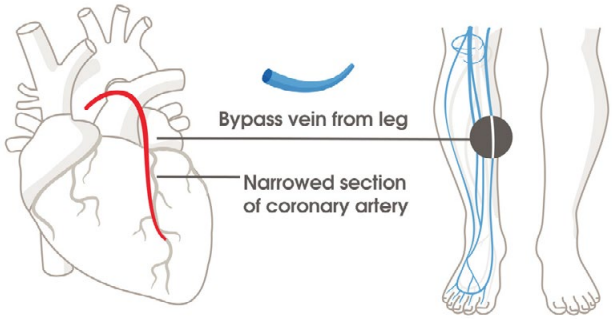
What is bypass surgery?

Bypass surgery is an operation in which a blood vessel is taken from your chest, leg or forearm and grafted (attached) to your coronary artery to let blood 'detour' past a narrowing or blockage in this artery. This type of surgery is used to improve blood flow to your heart muscle, when blood flow is reduced due to coronary heart disease (CHD) (see Coronary heart disease box on page 4).

Bypass surgery is a common and successful way to improve the health of people with CHD. This surgery usually takes between three and six hours.





Did you know?

The full name of bypass surgery is 'coronary artery bypass graft surgery'. It is shortened to 'CABG' and pronounced 'cabbage'.

Why do I need bypass surgery?

You need bypass surgery because one or more of your coronary arteries has become too narrow or blocked, and not enough blood can reach your heart muscle. This is probably because you have coronary heart disease.

Your doctor will discuss your treatment options with you. They will tell you if bypass surgery is the most suitable form of treatment to improve blood flow to your heart muscle and reduce your risk of further heart problems.

What are the benefits of bypass surgery?

Bypass surgery will improve blood flow to your heart muscle and reduce your symptoms of CHD, such as angina. In time, you may feel healthier and able to be more active than you were before the surgery. Your risk of further heart problems will also be reduced.

Coronary heart disease

Coronary heart disease is when your coronary arteries (the ones that carry oxygen and nutrients to your heart muscle) become clogged with fatty material called 'plaque' or 'atheroma'. Plaque slowly builds up on the inner wall of your arteries, causing them to become narrow. This process is called 'atherosclerosis'. It may start when you are young and be well advanced by middle age.

If your arteries become too narrow, the blood supply to your heart muscle is reduced. This may lead to symptoms such as angina.

If a blood clot forms in the narrowed artery and blocks the blood supply to a particular part of your heart, it can cause a life-threatening heart attack.

What will happen before I have bypass surgery?

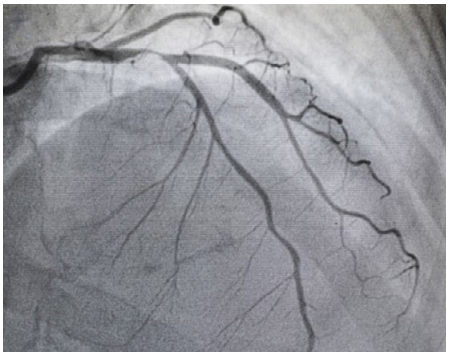
Preparing for bypass surgery

Coronary angiography

To find out if you need bypass surgery, your doctor will assess your symptoms. You will have a 'coronary angiogram', a special type of real-time X-ray that shows the areas of narrowing or blockage in your coronary arteries.

During the coronary angiogram, you are given a local anaesthetic, and then a catheter (a long thin tube) is put into an artery in your groin, arm or wrist. The catheter is moved up the inside of your artery until it reaches your heart. A special dye is then injected into your coronary arteries, and X-rays are taken.

The X-ray shows your doctor detailed information about your coronary arteries, including whether or not your arteries are narrowed and if so, how narrow they are.



Coronary angiogram

Want to know more?

For more information, call your GP. You can also visit heartfoundation.org.au

Talk to your doctor

Find out as much as you can about bypass surgery, and what to expect when you are in hospital and when you are recovering.

Talk to your doctor about any concerns that you have about bypass surgery. The more you understand, the less anxious you will feel.

If your hospital runs a before surgery 'briefing' or 'pre admission information session', try to attend. The session will give you information about bypass surgery and help you to plan for it.

Quit smoking

If you smoke, your doctor will most likely advise you to give up before bypass surgery. Quitting smoking will help your lungs to cope with surgery and reduce your risk of further health problems.

If you do smoke, there is a greater risk that your coronary artery disease will advance and may result in a heart attack or the need for further surgery.

Talk to your doctor or pharmacist about quit smoking support and nicotine replacement therapy to help you to be smoke-free before and after your surgery.

Medicines

Talk to your doctor about your medications and if any changes are required before the surgery. Make sure your medication list is up to date and you understand your doctor's instructions.

The day before bypass surgery

You will probably go to hospital the day before or on the morning of your bypass surgery, so that you can get ready and meet the medical team that will do your operation. The team may include surgeons, nurses, cardiologists, anaesthetists and therapists who will care for you before, during and after surgery.

The medical team will talk with you about your bypass surgery and recovery, and can answer any questions that you have.

The anaesthetist will ask you questions about your medical history and if you have any known allergies.

Much of your body hair will be clipped. Special attention will be paid to cleaning your chest, arms and legs. You may be asked to use a surgical antiseptic solution in the shower before the operation.

The day of bypass surgery

Give personal items, such as glasses, contact lenses, dentures, watches and jewellery, to family members or the nurse to look after before you go into the operating room.

About an hour before bypass surgery, you will be given medicine to make you sleepy and keep you relaxed. You will be asked to lie on a rolling bed and will be taken to the operating room. In the operating room, you will be given anaesthetic medications to keep you asleep, still and pain free during the surgery.

Most hospitals have a waiting area for families. If your family decides to wait at home, the hospital will call after surgery to let them know how it went. Make sure the hospital knows where your family is if they need to be contacted or updated.

Some hospitals may let close relatives briefly visit the recovery room or intensive care unit (ICU) shortly after bypass surgery, while you are still asleep.

What will happen when I have bypass surgery?

The surgeons will cut along the midline of your chest, through your breastbone (sternum), to reach your heart.

During surgery, you may be connected to a heart/lung machine, which will temporarily take over from your own heart and lungs during surgery.

Sometimes surgery can be performed through a smaller incision in your chest and without using a heart/lung machine. Whether or not this is possible depends on the number of narrowings and where they are in your coronary arteries.

If the blood vessel for the bypass graft is coming from your leg or arm, a cut will be made in the area from where the blood vessel will be removed. A cut may also be made in your groin so that the large blood vessels can be reached. The surgeon will insert the bypass graft before and after the narrowing in your artery which creates a new path for the blood to flow. The surgeon will identify all significant narrowings and make as many artery bypasses as required.

What are the risks of bypass surgery?

There are risks associated with all surgeries and often bypass surgery can take many hours and involve special procedures. Depending on your age, health conditions and other factors your risk of complications may be different. You will be asked to sign a consent form before the surgery and you will be encouraged to ask questions to help you understand your individual risk.

What will happen after I have bypass surgery?

After the operation, you will be taken to a recovery area or intensive care unit (ICU) for close supervision until you wake up from the anaesthetic. You will probably stay in the intensive care unit for one or two days. After the intensive care unit, you will be moved to another ward until you are ready to go home.

When you wake up, you will have tubes and wires attached to parts of your body.

You will have a breathing tube in your mouth that passes between your vocal cords and windpipe.

It isn't painful but will stop you talking. It can be uncomfortable but try to be patient. Your healthcare team will remove the tube as soon as it is safe to do so.

You may feel confused when you first wake up, and you may find it hard to keep track of time. This is normal and may be caused by the effects of painkilling medicines, the lights being on 24 hours a day and all of the activity going on around you.

Any confusion should only last a day or two until you are moved to a quieter ward.

Before you leave hospital, your doctor will make follow-up appointments for you to discuss your medicines, wound care and physical activity. Your cardiologist will continue to oversee your general heart health.

Your first outpatient appointment will usually be four to six weeks after you leave hospital. You will see your surgeon, or a member of your medical team. They will check your wound and general progress, and you may also have an X-ray at this time.

Will I feel pain after bypass surgery?

You will probably feel some soreness and discomfort, especially during the first few days after surgery. You will be given pain relievers regularly to make sure that you don't feel severe pain.

If the pain starts to build up, tell your nurse. You and your nurse need to communicate and manage your pain together.

Medicine won't get rid of all of your pain, but it can manage it.

Stronger pain relievers are needed for only a few days, so there is very little risk of becoming addicted to them.

How long will I have to stay in hospital?

This varies depending on your condition and your response to surgery. On average, after bypass surgery, people stay in hospital for four to seven days.

How long will it take to recover from bypass surgery?

Normal recovery from heart surgery takes four to six weeks. During this time, you will begin to strengthen your arm and leg muscles and return to your usual activities.

Usually wire or special strips of tape are used to hold your breastbone together. These do not need to be removed. It takes about six to 12 weeks for your breastbone to heal completely.

During that time you shouldn't lift anything heavy.

Soon after your operation, your chest wound will be exposed to the air, allowing it to dry. After a few days, the wound can be cleaned.

If the mammary artery was used in the graft, you may feel numbness or a different or unusual sensation down the side of your breastbone. This usually changes to a super-sensitive feeling after a few weeks and gradually returns to normal.

If the graft was taken from your leg, you may feel numbness around your ankle. This numbness may last for several weeks.

Your ankle may swell, but wearing elastic support stockings and putting your legs up when you rest can help to relieve this.

Walking will help your leg to recover and is good for your heart, but ask your doctor when you can start being more active.

If the graft was taken from your forearm, you may feel numbness on the back of your hand at the base of the thumb. This numbness may also last for several weeks before returning to normal.

After major surgery, such as bypass surgery, recovery time may seem to pass slowly. Your body has been slowed down by lowered activity, lack of good sleep, medicines and the surgery itself. You may feel drained physically and emotionally. Many patients feel 'down' for the first fortnight after their surgery and this is considered a normal experience. You should start to feel better in the first month after your surgery. If you are concerned that your mood and emotions are not improving, see your doctor.

Quicktip: It's important that you are free of pain when you go to bed so that you get a good night's sleep. At first it will be most comfortable to lie on your back, propped up at about 45 degrees. In time, you will be able to sleep in your normal position again. You might also find it helpful to learn relaxation techniques or listen to relaxing music to help you to sleep.

What can I do to help me to recover more quickly?

You should aim to get out of bed on the first day after your surgery with the help of nurses and physiotherapists. Sitting up and assisted walking as soon as possible will help you to recover more quickly and will reduce your risk of chest infection.

An exercise routine moving your arms and shoulders can help reduce upper body stiffness and soreness.

While they may cause some discomfort to start with, they won't do any damage and will help your recovery. Ask your nurse for pain relief before you exercise so that your pain does not restrict your movement. It is important that when you cough, you hug a towel or a pillow across your chest to support your breastbone, reduce pain and to assist with wound healing.

Your physiotherapist will assist you with developing a walking program, exercise program, and to show you the correct techniques to clear your phlegm if necessary.

Your health care team should refer you to cardiac rehabilitation for you to attend after you go home. Cardiac rehabilitation is a structured education and exercise program that usually runs for six to eight weeks. Cardiac rehabilitation will help you recover, return to physical activity, and give you opportunity to ask questions and interact with others in a similar situation.

Cardiac rehabilitation

The Heart Foundation and World Health Organization recommend that all patients who have had a heart attack, heart surgery, coronary angioplasty, angina or other heart and blood vessel disease attend an appropriate cardiac rehabilitation and prevention program.

It is highly recommended that you participate in cardiac rehabilitation and talk to your health care team about a referral to a service that suits you, which may be face to face, online or even over the phone.

When can I start eating again?

After your breathing tube is removed, you will be able to swallow a small amount of liquid and build up to foods over the next couple of days.

How will I feel about leaving hospital?

Many people feel a little anxious and depressed about leaving the security of hospital to go home. Remember, you will only be allowed to leave if the doctor thinks that you are well enough and will continue to recover at home.

What happens when I get home?

For the first few weeks, your family and friends will need to keep a close eye on you and help you.

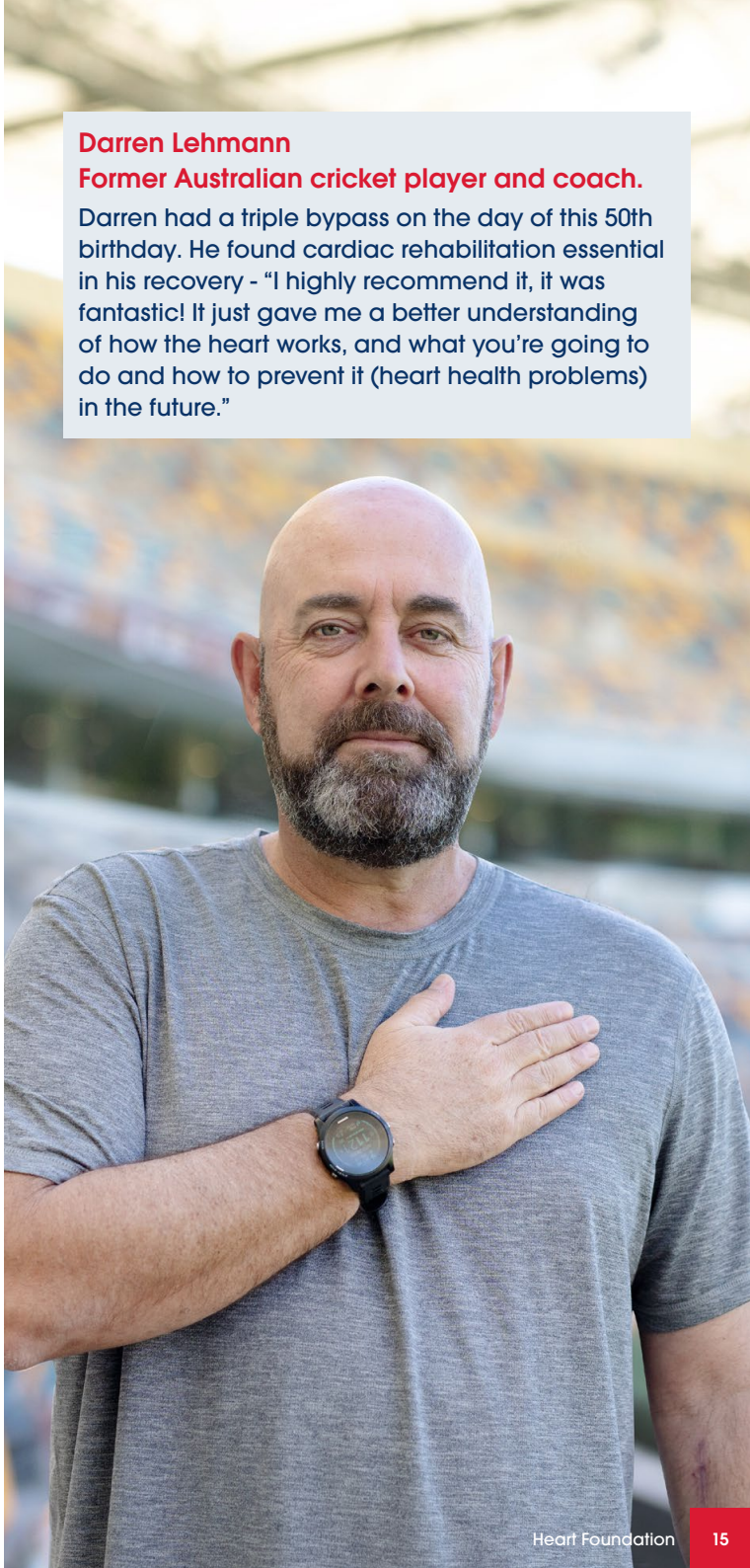
If you normally live on your own, someone will need to stay with you for the first few weeks after you return home. Some home care agencies can help you until you can manage on your own again. If you need extra community services, talk to your doctor or nurse.

You will need lots of rest, but try to get back into a normal routine as soon as possible.

- Get up at a normal hour.
- Bathe or shower if possible.
- Get dressed for the day: don't stay in your sleeping clothes during the day.
- Women should wear a wireless bra to support their chest and reduce discomfort.
- Rest mid-morning, mid-afternoon and after physical activity.

Darren Lehmann
Former Australian cricket player and coach.

Darren had a triple bypass on the day of this 50th birthday. He found cardiac rehabilitation essential in his recovery - "I highly recommend it, it was fantastic! It just gave me a better understanding of how the heart works, and what you're going to do and how to prevent it (heart health problems) in the future."



You should be able to:

- help with light work around the house
- go to places, such as a theatre, restaurant or church
- visit friends
- ride in a car
- climb stairs slowly.

You may feel weak when you first go home, but this isn't usually serious. It is normally because you didn't use your muscles very much while you were in hospital.

What should I be eating?

Enjoying a variety of foods from the different food groups is the key to healthy eating. Choose:

- mainly plant-based foods, such as vegetables, fruit, legumes (e.g. chick peas, kidney beans, baked beans and lentils) and plain unsalted nuts
- wholegrain cereal foods, such as breakfast cereals, bread, pasta, noodles and rice
- moderate amounts of lean unprocessed meats, skinless poultry and reduced, low or no fat dairy products. Avoid more than 350 grams of red meat a week
- oily fish – include a 150 gram serve in two or three meals per week
- foods with low or reduced salt content and foods labelled 'no added salt', 'low salt' or 'salt reduced'
- plant-based oils, such as canola, sunflower, soybean, olive and peanut oils.

Your doctor, dietitian or nursing staff will give you advice on healthy eating. Visit the Heart Foundation website for heart healthy recipes and ideas

When can I be more physically active?

Once you are at home, continue with the level of activities that you started in hospital. Your surgical or rehabilitation team will give you advice on what physical activity to do over the coming weeks and months. They will probably create a physical activity plan for your needs.

Set realistic goals for yourself. Don't over do it, but don't be inactive either. Rest when you need to and change an activity if it is making you too tired. Doing too much at this time won't injure the heart. It will, however, make you very tired.

Remember to start slowly. For the first few weeks after surgery, you will need lots of rest, as well as help to do the shopping, cooking and heavy household tasks.

Walking is a very good physical activity after bypass surgery. Gradually build up how long and how quickly you walk, but don't do more than your doctor recommends.

As you become more active during your recovery, you will probably get some pain associated with the wound or muscles in your chest, shoulder, neck and back. Continue to take your pain-relieving tablets (e.g. paracetamol) when you need them.

You will be able to reduce these gradually. **Don't neglect any type of severe pain**.

Important note about physical activity

If you experience chest wound complications, such as redness or weeping and/or movement of the breastbone, contact your surgical team for advice. You may need to stop your activity or take it a bit more slowly and allow more time for healing.

When can I have sex?

Everyone's recovery time is different. When you feel that you have recovered enough, which you can judge from how much pain you have and the level of physical activity that you can do, you should be able to have sex again. For example, if you can climb two flight of stairs without getting chest pain or shortness of breath, you will probably be able to cope with the amount of energy required for sex.

Be careful not to put any pressure or stress on your chest. You may find that some positions are more comfortable than others, because of discomfort in your chest wound. It can take time for sexual relationships to get back to normal after surgery, so try to take things slowly. Make an effort to discuss your feelings and concerns with your partner.

Remember to stop any activity, including sex, if you feel pain or discomfort in your chest.

When can I drive, and can I wear a seatbelt?

Your doctor and/or rehabilitation team will tell you when you can start driving again. You may have to avoid driving a car for one to two months after bypass surgery because your reaction time will be slower due to weakness, tiredness and medicines.

Contact the relevant authority in your state, and/or your insurance company, to seek advice on any restrictions or implications for your insurance coverage.

You can wear a seatbelt as usual. If you feel pain from wearing a seatbelt, try wearing a jumper for extra protection.

What if I have to travel on public transport?

If you have to travel by train, bus or plane, let the transport company know about your condition and they will usually let you board before other passengers.

If you need a wheelchair, let the transport company know and they will make arrangements for you.

When can I return to work?

This decision is usually made after your four to six week checkup.

It will depend on your type of work, how demanding your job is, your strength and other factors. Office workers can usually return to work in six weeks. If your work is more physically demanding, follow your doctor's advice.

Why do I feel great one day, but dreadful the next?

It is normal to feel 'down', frustrated, anxious, angry, worried or tired after bypass surgery. These feelings often do not last long and may be eased by talking to the people around you, such as your doctor, cardiac rehabilitation team, family and friends.

Doing something interesting each day and not changing your normal lifestyle too much can also help.

While negative feelings are common and normal after bypass surgery, symptoms lasting longer than a couple of weeks may indicate depression.

Depression is more common in people with coronary heart disease than in the general population.

People with depression usually feel sad, down or miserable most of the time, and find it hard to do normal activities. If you are experiencing symptoms of depression for more than two weeks, it is very important that you talk to your doctor. They can discuss with you treatments that can help.

Remember

Call your doctor if:

- your pain is severe or gets worse
- you have any sign of infection, fever, chills or swelling
- you are short of breath
- you are putting on a lot of weight
- there is a change in the speed or rhythm of your heartbeat
- there are any other symptoms that worry you.

Will I need to take medicines?

After bypass surgery, you will need to take medicines.

However, only take medicines prescribed by your doctor.

You must tell your doctor about all of the medicines you were taking before your operation, including over-the-counter (non-prescription) medicines and supplements. Don't keep taking them after your operation unless you have talked to your doctor about them and he or she prescribes them for you.

Your doctor will monitor your blood pressure and cholesterol.

He or she is likely to prescribe a cholesterol-lowering medicine, such as a statin, even if your cholesterol is in the 'normal' range.

You might also be prescribed blood thinner medicines, such as aspirin or clopidogrel, or other medicines that you need to take for the long term.

It is important to take all of your medicines for as long as your doctor tells you to.

Remember

If you have any concerns about side effects, tell your doctor. It is important that you work closely with your doctor to find the medicine (or medicines) that work best for you. Your local pharmacist is also a great source of information and support about your medicines.

How can I reduce my risk of further heart problems?

The things that you do and the choices you make can greatly reduce your risk of more heart problems in the future, including heart attack.

The most important things you can do to reduce your risk of more heart problems are to:

- take your medicines as prescribed by your doctor
- be smoke-free
- enjoy healthy eating
- limit your alcohol intake
- be physically active
- manage your blood pressure and cholesterol
- achieve and maintain a healthy body weight
- maintain your psychological and social health.

People with diabetes should generally aim to maintain their blood glucose levels within the normal non-diabetic range and follow individual advice from their doctor or accredited diabetes educator.

Heart attack

A heart attack is an emergency. Getting to hospital quickly can reduce the damage to your heart and increase your chance of survival.

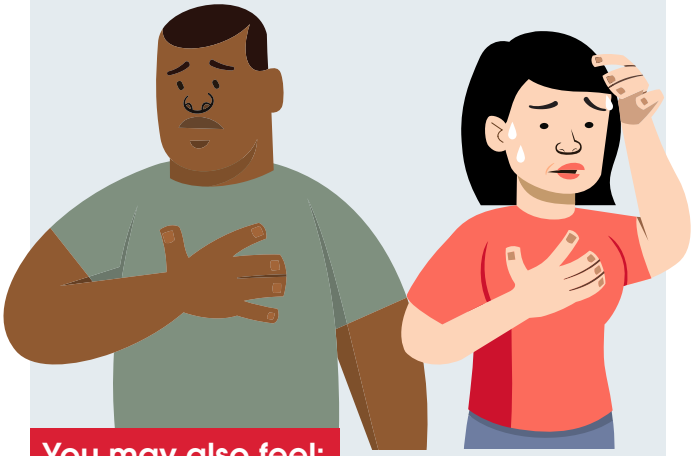
A heart attack happens when there is a sudden, blockage of an artery that supplies blood to your heart muscle (a 'coronary artery'). As a result, some of your heart muscle starts to die. If the artery stays blocked, the lack of blood permanently damages the area of your heart muscle that is supplied by that artery.

It is important that you know what to do if you or someone else has a heart attack

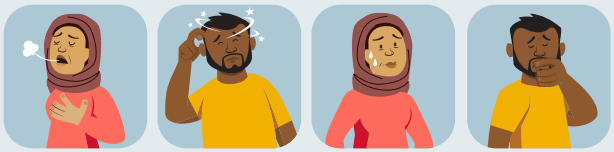
Warning signs of a heart attack

Pain, pressure or tightness in one or more of these areas:

Chest | Arm/s | Shoulder/s or Back | Neck or Jaw



You may also feel:



Short of breath | Dizzy | Sweaty | Sick

Tell someone how you feel.
If feeling worse or not better after 10 minutes,

Call Triple Zero (000)

Take 300mg of aspirin if you have it, unless you are allergic or your doctor has told you not to.

What should I talk to my doctor about?

Bypass surgery is a major event in your life and you should make sure that you are well informed about why you need the surgery, and what the benefits and risks of surgery are.

Below are some questions about bypass surgery that you may want to ask your doctor.

1. **What benefits will I get from surgery?**
2. **What are the risks or potential complications of surgery?**
3. **Will I need a blood transfusion? Can I donate blood for myself?**
4. **How many bypasses will I have?**
5. **Will there be a lot of pain?**
6. **What medicines will I need to take after surgery?**
7. **Will I live longer if I have surgery?**
8. **How much will surgery cost and how much will be covered by insurance?**
9. **How long will I be in hospital?**
10. **If I need to travel from my home to a different town for surgery, how long should I stay there after I'm discharged from hospital?**
11. **Is there a place nearby where my partner can stay while I'm in hospital?**
12. **How soon after surgery can I go back to work?**
13. **How often should I have follow-up visits?**

Key things to remember about bypass

Bypass surgery is a common and successful way to improve the health of people with coronary heart disease.

The full name of bypass surgery is 'coronary artery bypass graft surgery'. It is shortened to 'CABG' and pronounced 'cabbage'.

After bypass surgery, you will need lots of rest, but try to get back into a normal routine as soon as possible.

After bypass surgery, you will need to take medicines. However, only take medicines prescribed by your doctor. It is important to take all of your medicines for as long as your doctor tells you to. If you have any concerns about side effects, tell your doctor. The most important things you can do to reduce your risk of more heart problems are to:

- take your medicines as prescribed by your doctor
- be smoke-free
- enjoy healthy eating
- limit your alcohol intake
- be physically active
- manage your blood pressure and cholesterol
- achieve and maintain a healthy bodyweight
- maintain your psychological and social health.

Heart Foundation

For heart health information visit **heartfoundation.org.au**

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